

PS1 - IO

Tate Mason

```
# Packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(lubridate)

# Data
iri <- read_tsv("~/SchoolWork/Y2S1/IO/PS1/IRI.csv", show_col_types = FALSE)
names(iri) <- gsub("^t", "", names(iri)) # Clean names
iri <- iri %>%
  mutate(
    market_id = interaction(store_id, week_id, drop = TRUE),
    date_id = interaction(year, month, drop = TRUE),
    date = as.Date(paste(year, month, "01", sep = "-"))
  )
```

Question 1:

1.1

```
# Distinct counts per market
brands_per_market <- iri %>%
  distinct(market_id, brand) %>%
  count(market_id, name = "n_brands")

mfrs_per_market <- iri %>%
  distinct(market_id, parent) %>%
  count(market_id, name = "n_mfrs")

# Total market sales per market
total_sales_market <- iri %>%
  group_by(market_id) %>%
  summarise(total_sales = sum(quantity, na.rm = TRUE), .groups = "drop")

# Brand-level sales (market x brand)
brand_sales_market <- iri %>%
  group_by(market_id, brand) %>%
  summarise(brand_sales = sum(quantity, na.rm = TRUE), .groups = "drop")

# Per-market price & characteristics (means within market)
market_chars <- iri %>%
  group_by(market_id) %>%
  summarise(
    price_mean      = mean(price, na.rm = TRUE),
    fiber_mean      = mean(fiber, na.rm = TRUE),
    sugars_mean     = mean(sugars, na.rm = TRUE),
    flavored_mean   = mean(as.numeric(flavored), na.rm = TRUE),
    fortified_mean  = mean(as.numeric(fortified), na.rm = TRUE),
    .groups = "drop"
  )

# Helper to summarise numeric vectors
summarise_vec <- function(x) {
  tibble(
    mean  = mean(x, na.rm = TRUE),
    sd    = sd(x, na.rm = TRUE),
    median = median(x, na.rm = TRUE),
```

```

    max      = max(x, na.rm = TRUE)
  )
}

# Tables for 1.1
tbl_counts <- bind_cols(
  variable = c("n_brands", "n_mfrs"),
  bind_rows(
    summarise_vec(brands_per_market$n_brands),
    summarise_vec(mfrs_per_market$n_mfrs)
  )
)

tbl_total_sales <- bind_cols(
  variable = "total_sales (per market)",
  summarise_vec(total_sales_market$total_sales)
)

tbl_brand_sales <- bind_cols(
  variable = "brand_sales (per market-brand)",
  summarise_vec(brand_sales_market$brand_sales)
)

tbl_price_chars <- tribble(
  ~variable,          ~mean, ~sd, ~median, ~max,
  "price (market mean)",
    summarise_vec(market_chars$price_mean)$mean,
    summarise_vec(market_chars$price_mean)$sd,
    summarise_vec(market_chars$price_mean)$median,
    summarise_vec(market_chars$price_mean)$max,
  "fiber (market mean)",
    summarise_vec(market_chars$fiber_mean)$mean,
    summarise_vec(market_chars$fiber_mean)$sd,
    summarise_vec(market_chars$fiber_mean)$median,
    summarise_vec(market_chars$fiber_mean)$max,
  "sugars (market mean)",
    summarise_vec(market_chars$sugars_mean)$mean,
    summarise_vec(market_chars$sugars_mean)$sd,
    summarise_vec(market_chars$sugars_mean)$median,
    summarise_vec(market_chars$sugars_mean)$max,
  "flavored share (market mean)",
    summarise_vec(market_chars$flavored_mean)$mean,

```

```

    summarise_vec(market_chars$flavored_mean)$sd,
    summarise_vec(market_chars$flavored_mean)$median,
    summarise_vec(market_chars$flavored_mean)$max,
    "fortified share (market mean)",
    summarise_vec(market_chars$fortified_mean)$mean,
    summarise_vec(market_chars$fortified_mean)$sd,
    summarise_vec(market_chars$fortified_mean)$median,
    summarise_vec(market_chars$fortified_mean)$max
  )

summary_11 <- bind_rows(
  tbl_counts,
  tbl_total_sales,
  tbl_brand_sales,
  tbl_price_chars
)

summary_11

```

```

# A tibble: 9 x 5
  variable                mean      sd   median     max
  <chr>                  <dbl>   <dbl>   <dbl>   <dbl>
1 n_brands                34.9    4.82    37      39
2 n_mfrs                   3.95    0.259    4       4
3 total_sales (per market) 15555.  12358.  12557.  84567.
4 brand_sales (per market-brand) 446.    726.    238.   26470.
5 price (market mean)      0.235    0.0265  0.233    0.325
6 fiber (market mean)      1.30     0.0874  1.31     1.67
7 sugars (market mean)     8.31     0.205   8.33     10.2
8 flavored share (market mean) 0.640    0.0265  0.641    0.857
9 fortified share (market mean) 0.485    0.0341  0.487    0.667

```

```

# If you want a file:
# write_csv(summary_11, "summary_11_descriptives.csv")

```

1.2

```

ts_simple <- iri %>%
  group_by(week_id) %>%

```

```

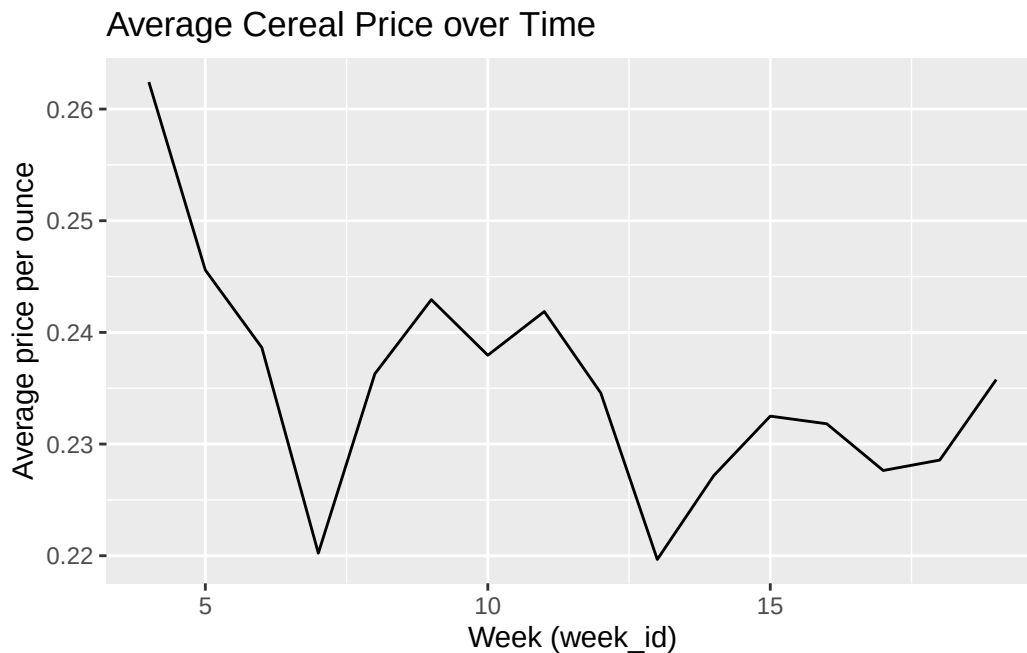
summarise(
  avg_price = mean(price, na.rm = TRUE),
  avg_qty   = mean(quantity, na.rm = TRUE),
  .groups = "drop"
) %>%
arrange(week_id)

p_price <- ggplot(ts_simple, aes(x = week_id, y = avg_price)) +
  geom_line() +
  labs(title = "Average Cereal Price over Time",
       x = "Week (week_id)", y = "Average price per ounce")

p_qty <- ggplot(ts_simple, aes(x = week_id, y = avg_qty)) +
  geom_line() +
  labs(title = "Average Cereal Sales (Quantity) over Time",
       x = "Week (week_id)", y = "Average quantity (lbs)")

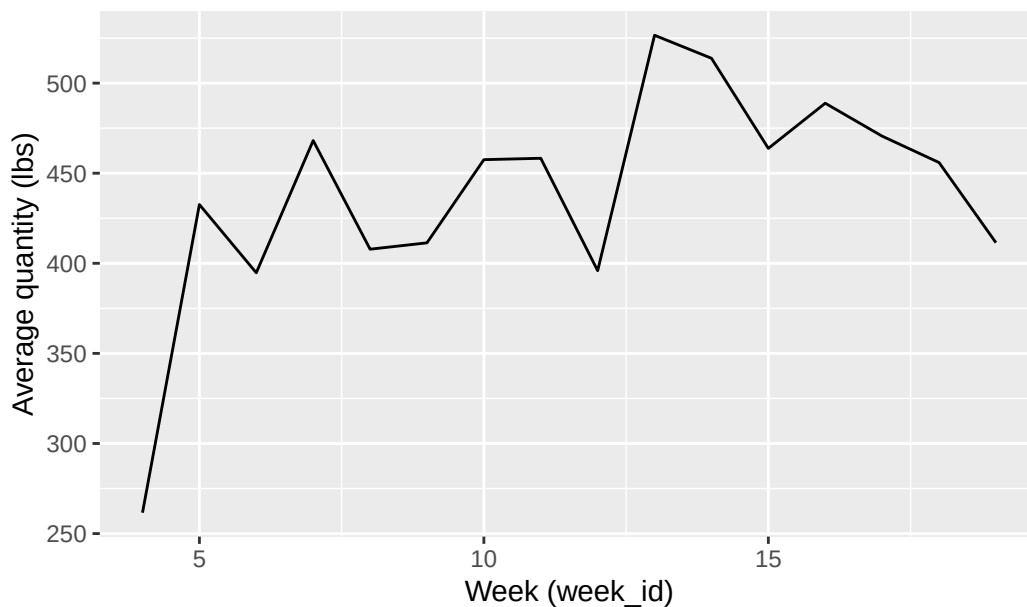
p_price

```



```
p_qty
```

Average Cereal Sales (Quantity) over Time

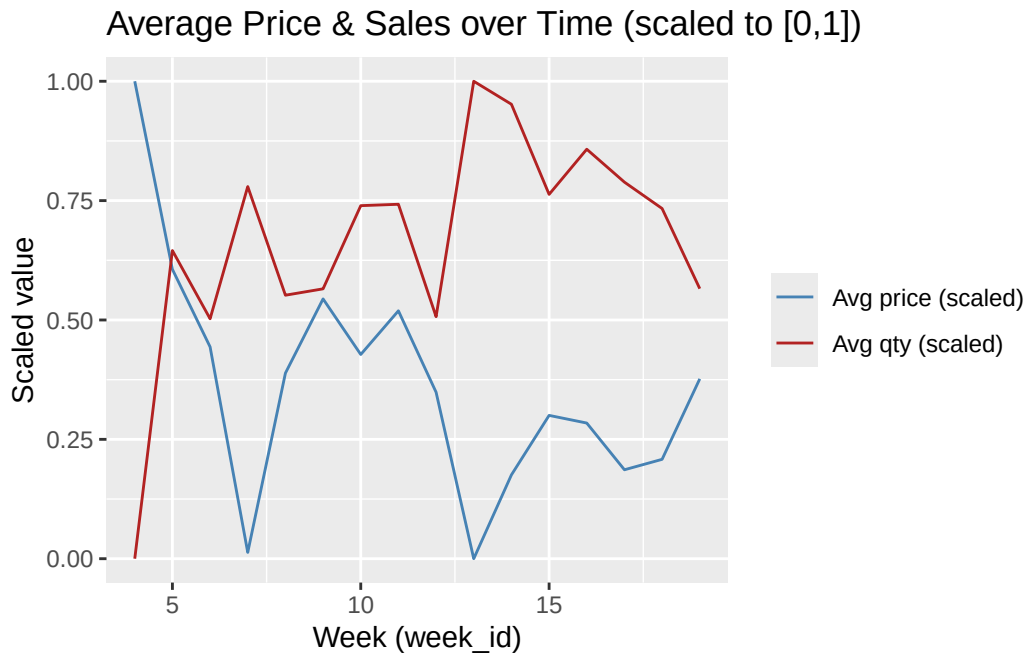


```
# Save figures
ggsave("avg_price.png", p_price, width = 10, height = 6, dpi = 300)
ggsave("avg_sales.png", p_qty, width = 10, height = 6, dpi = 300)

# Dual-axis style (scaled)
ts_long_scaled <- ts_simple %>%
  mutate(
    price_scaled = (avg_price - min(avg_price, na.rm = TRUE)) /
      (max(avg_price, na.rm = TRUE) - min(avg_price, na.rm = TRUE)),
    qty_scaled = (avg_qty - min(avg_qty, na.rm = TRUE)) /
      (max(avg_qty, na.rm = TRUE) - min(avg_qty, na.rm = TRUE))
  ) %>%
  select(week_id, price_scaled, qty_scaled) %>%
  pivot_longer(-week_id, names_to = "series", values_to = "value")

p_dual <- ggplot(ts_long_scaled, aes(x = week_id, y = value, color = series)) +
  geom_line() +
  scale_color_manual(values = c("price_scaled" = "steelblue", "qty_scaled" = "firebrick"),
    labels = c("Avg price (scaled)", "Avg qty (scaled)")) +
  labs(title = "Average Price & Sales over Time (scaled to [0,1])",
    x = "Week (week_id)", y = "Scaled value", color = NULL)

p_dual
```



```
ggsave("avg_price_sales_dual.png", p_dual, width = 10, height = 6, dpi = 300)
```

1.3

```
# Brand HHI per market
brand_shares <- iri %>%
  group_by(market_id) %>%
  mutate(total_mkt_sales = sum(quantity, na.rm = TRUE)) %>%
  ungroup() %>%
  group_by(market_id, brand) %>%
  summarise(brand_sales = sum(quantity, na.rm = TRUE),
            total_mkt_sales = first(total_mkt_sales),
            .groups = "drop") %>%
  mutate(share_brand = if_else(total_mkt_sales > 0, brand_sales / total_mkt_sales, 0))

hhi_brand_market <- brand_shares %>%
  group_by(market_id) %>%
  summarise(HHI_brand = sum(share_brand^2, na.rm = TRUE), .groups = "drop")

hhi_brand_week <- iri %>%
  distinct(market_id, week_id) %>%
```

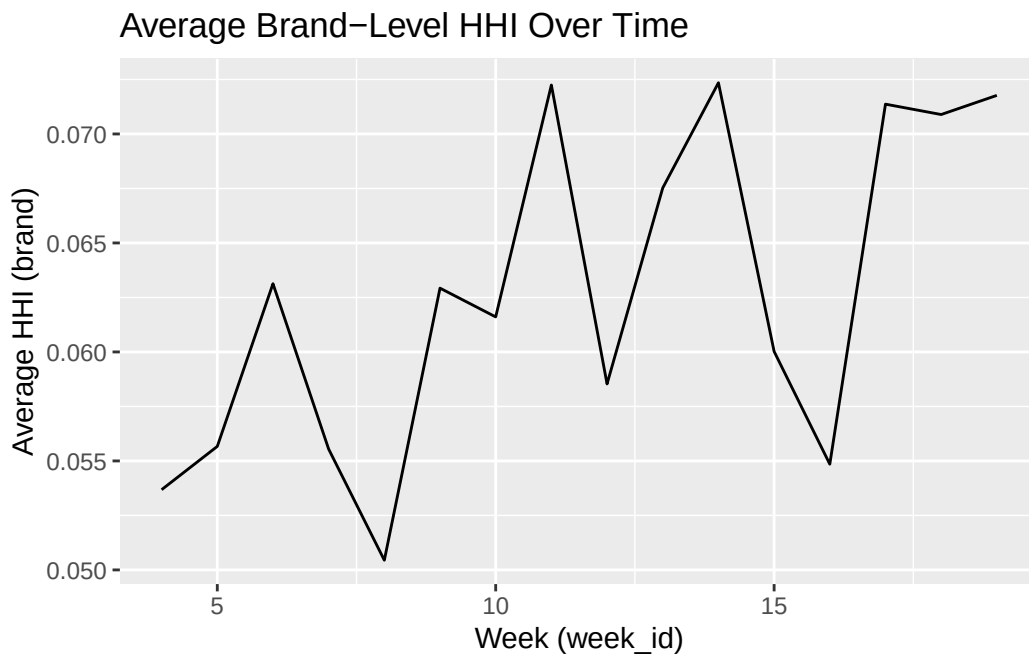
```

inner_join(hhi_brand_market, by = "market_id") %>%
group_by(week_id) %>%
summarise(avg_HHI_brand = mean(HHI_brand, na.rm = TRUE), .groups = "drop") %>%
arrange(week_id)

p_hhi_brand <- ggplot(hhi_brand_week, aes(x = week_id, y = avg_HHI_brand)) +
  geom_line() +
  labs(title = "Average Brand-Level HHI Over Time",
       x = "Week (week_id)", y = "Average HHI (brand)")

p_hhi_brand

```



```

ggsave("HHI_brand.png", p_hhi_brand, width = 10, height = 6, dpi = 300)

# Manufacturer (parent) HHI per market
parent_shares <- iri %>%
  group_by(market_id) %>%
  mutate(total_mkt_sales = sum(quantity, na.rm = TRUE)) %>%
  ungroup() %>%
  group_by(market_id, parent) %>%
  summarise(parent_sales = sum(quantity, na.rm = TRUE),
            total_mkt_sales = first(total_mkt_sales),
            .groups = "drop") %>%

```

```

mutate(share_parent = if_else(total_mkt_sales > 0, parent_sales / total_mkt_sales, 0))

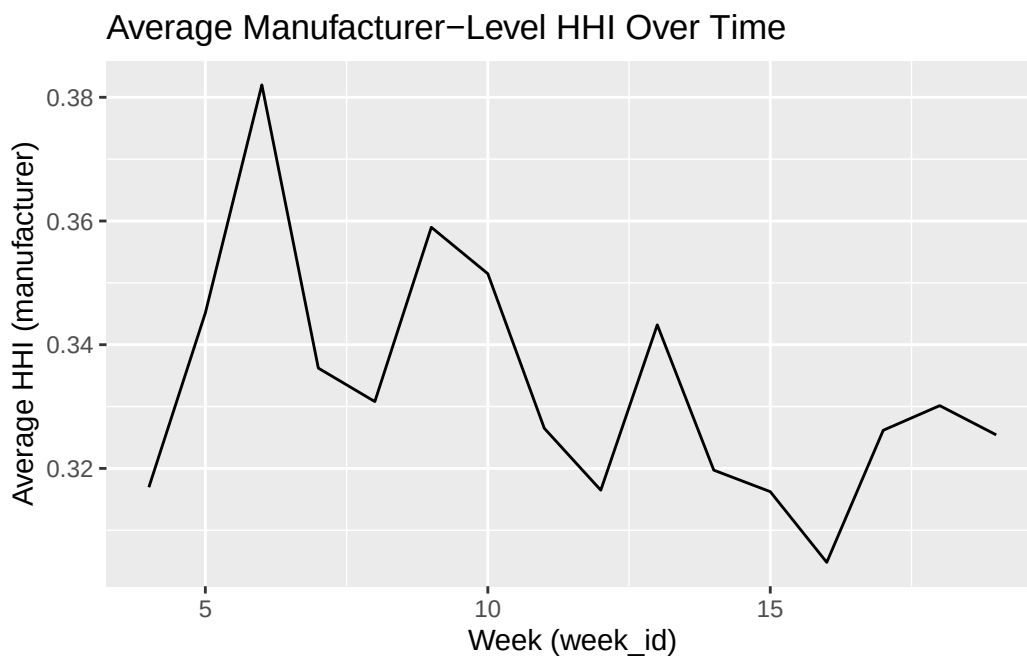
hhi_parent_market <- parent_shares %>%
  group_by(market_id) %>%
  summarise(HHI_parent = sum(share_parent^2, na.rm = TRUE), .groups = "drop")

hhi_parent_week <- iri %>%
  distinct(market_id, week_id) %>%
  inner_join(hhi_parent_market, by = "market_id") %>%
  group_by(week_id) %>%
  summarise(avg_HHI_parent = mean(HHI_parent, na.rm = TRUE), .groups = "drop") %>%
  arrange(week_id)

p_hhi_parent <- ggplot(hhi_parent_week, aes(x = week_id, y = avg_HHI_parent)) +
  geom_line() +
  labs(title = "Average Manufacturer-Level HHI Over Time",
       x = "Week (week_id)", y = "Average HHI (manufacturer)")

p_hhi_parent

```



```

ggsave("HHI_parent.png", p_hhi_parent, width = 10, height = 6, dpi = 300)

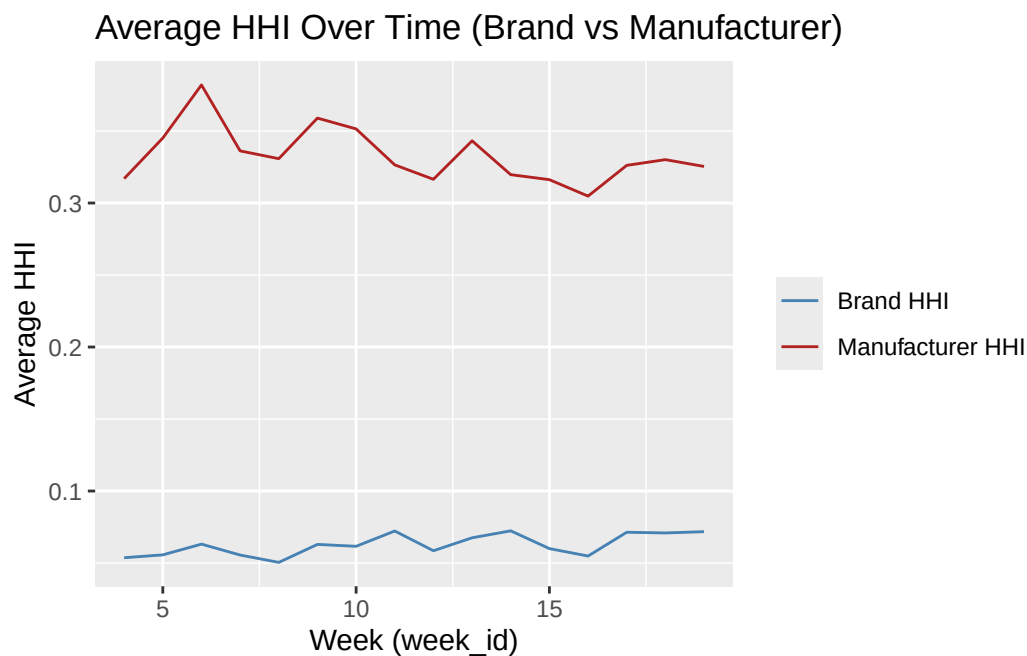
# Combined comparison

```

```
hhi_both <- hhi_brand_week %>% inner_join(hhi_parent_week, by = "week_id")

p_hhi_both <- ggplot(hhi_both, aes(x = week_id)) +
  geom_line(aes(y = avg_HHI_brand, color = "Brand HHI")) +
  geom_line(aes(y = avg_HHI_parent, color = "Manufacturer HHI")) +
  scale_color_manual(values = c("Brand HHI" = "steelblue", "Manufacturer HHI" = "firebrick")) +
  labs(title = "Average HHI Over Time (Brand vs Manufacturer)",
       x = "Week (week_id)", y = "Average HHI", color = NULL)

p_hhi_both
```



```
ggsave("HHI_brand_vs_parent.png", p_hhi_both, width = 10, height = 6, dpi = 300)
```

Question 2:

2.1

```
# Prepare data for regression
reg_data <- iri %>%
  group_by(market_id) %>%
```

```
mutate(
  market_fe = as.factor(state),
  manufacturer_fe = as.factor(parent),
  time_fe = as.factor(week_id),
  brand_fe = as.factor(brand),
  p_jmt = price,
  total_sales = sum(quantity, na.rm = TRUE),
  sj = quantity / M,
  s0 = 1 - (total_sales / M),
  u = log(sj) - log(s0)
) %>%
ungroup()
```

Our estimation model is thus:

$$u_{jmt} = \alpha p_{jmt} + \beta \mathbb{X}_{jmt} + \xi_{jmt}$$

where $\mathbb{X}_{jmt}$ includes the product characteristics fiber, sugars, flavored, and fortified, as well as fixed effects for market, manufacturer, and time. and u_{jmt} is the log-odds of product j's market share in market m at time t.

2.2

```
# Regression model
reg_model <- lm(u ~ p_jmt + fiber + sugars + flavored + fortified +
               market_fe + manufacturer_fe + time_fe + brand_fe,
               data = reg_data)
summary(reg_model)
```

Call:

```
lm(formula = u ~ p_jmt + fiber + sugars + flavored + fortified +
    market_fe + manufacturer_fe + time_fe + brand_fe, data = reg_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.0934	-0.4252	0.0600	0.4958	3.7968

Coefficients: (7 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.4054702	0.0638422	-6.351	2.15e-10
p_jmt	-6.6389158	0.0802566	-82.721	< 2e-16
fiber	-0.9442654	0.0164697	-57.333	< 2e-16
sugars	-0.4002088	0.0054006	-74.105	< 2e-16
flavored	0.7597779	0.0191629	39.648	< 2e-16
fortified	0.7032186	0.0181086	38.833	< 2e-16
market_feMassachusetts	-0.1051903	0.0084919	-12.387	< 2e-16
market_feTexas	-0.1519766	0.0103205	-14.726	< 2e-16
market_feIllinois	-0.0375639	0.0100223	-3.748	0.000178
market_feIndiana	-0.4650618	0.0264689	-17.570	< 2e-16
manufacturer_feKELLOGGS	3.3521849	0.0617187	54.314	< 2e-16
manufacturer_fePEPSICO INC	0.7512321	0.0230711	32.562	< 2e-16
manufacturer_feRALCORP HOLDINGS	1.4637079	0.0296752	49.324	< 2e-16
time_fe5	-0.0001525	0.0247749	-0.006	0.995090
time_fe6	-0.1727962	0.0231846	-7.453	9.23e-14
time_fe7	-0.0941736	0.0261641	-3.599	0.000319
time_fe8	0.0617927	0.0262584	2.353	0.018613
time_fe9	-0.0859918	0.0238900	-3.599	0.000319
time_fe10	-0.0888300	0.0230981	-3.846	0.000120
time_fe11	-0.1818979	0.0238172	-7.637	2.25e-14
time_fe12	-0.2368606	0.0236620	-10.010	< 2e-16
time_fe13	-0.2548369	0.0245400	-10.385	< 2e-16
time_fe14	-0.2969521	0.0232892	-12.751	< 2e-16
time_fe15	-0.2451704	0.0237494	-10.323	< 2e-16
time_fe16	-0.3108623	0.0257226	-12.085	< 2e-16
time_fe17	-0.3356226	0.0238612	-14.066	< 2e-16
time_fe18	-0.4433923	0.0233000	-19.030	< 2e-16
time_fe19	-0.3943541	0.0231974	-17.000	< 2e-16
brand_feGENERAL MILLS HONEY NUT CHEER	3.2169077	0.0345498	93.109	< 2e-16
brand_feGENERAL MILLS KIX	-1.4719961	0.0482058	-30.536	< 2e-16
brand_feGENERAL MILLS LUCKY CHARMS	1.0693120	0.0318262	33.598	< 2e-16
brand_feGENERAL MILLS MULTI GRAIN CHE	1.3007579	0.0295959	43.951	< 2e-16
brand_feGENERAL MILLS REESES PUFFS	0.7201245	0.0354311	20.325	< 2e-16
brand_feGENERAL MILLS TOTAL	1.7422502	0.0512161	34.018	< 2e-16
brand_feKELLOGGS APPLE JACKS	-0.6250416	0.0376848	-16.586	< 2e-16
brand_feKELLOGGS CORN FLAKES	-4.1907697	0.0915611	-45.770	< 2e-16
brand_feKELLOGGS CORN POPS	-0.7650356	0.0514965	-14.856	< 2e-16
brand_feKELLOGGS FROOT LOOPS	-1.4175250	0.0403043	-35.171	< 2e-16
brand_feKELLOGGS FROSTED FLAKES	-0.8320666	0.0513888	-16.192	< 2e-16
brand_feKELLOGGS FROSTED MINI WHEATS	-0.5542465	0.0349417	-15.862	< 2e-16
brand_feKELLOGGS RAISIN BRAN	1.1400337	0.0329477	34.601	< 2e-16
brand_feKELLOGGS RICE KRISPIES	-5.9856226	0.0943644	-63.431	< 2e-16

brand_feKELLOGGS SPECIAL K	-4.3832269	0.0807080	-54.310	< 2e-16
brand_feKELLOGGS SPECIAL K RED BERRIE	-3.4856503	0.0875502	-39.813	< 2e-16
brand_fePOST COCOA PEBBLES	-0.9864852	0.0403148	-24.470	< 2e-16
brand_fePOST FRUITY PEBBLES	-0.5247379	0.0401997	-13.053	< 2e-16
brand_fePOST GRAPE NUTS	-0.5977234	0.0277653	-21.528	< 2e-16
brand_fePOST HONEY BUNCHES OF OATS	0.1962830	0.0393273	4.991	6.02e-07
brand_feQUAKER CAP N CRUNCH	0.6339845	0.0303206	20.909	< 2e-16
brand_feQUAKER OATMEAL SQUARES	1.3757840	0.0292445	47.044	< 2e-16
brand_feGENERAL MILLS COCOA PUFFS	0.7582075	0.0354349	21.397	< 2e-16
brand_feGENERAL MILLS GOLDEN GRAHAMS	-1.2770623	0.0268852	-47.501	< 2e-16
brand_feGENERAL MILLS TOTAL RAISIN BR	1.6257957	0.0487126	33.375	< 2e-16
brand_feGENERAL MILLS TRIX	-0.0318812	0.0322734	-0.988	0.323231
brand_feKASHI GO LEAN CRUNCH	-3.0007301	0.0552377	-54.324	< 2e-16
brand_feKELLOGGS COCOA KRISPIES	-6.8814897	0.0940775	-73.147	< 2e-16
brand_feKELLOGGS RAISIN BRAN CRUNCH	NA	NA	NA	NA
brand_fePOST HONEYCOMB	-0.8470415	0.0409424	-20.689	< 2e-16
brand_fePOST RAISIN BRAN	2.4350469	0.0713761	34.116	< 2e-16
brand_fePOST SELECTS GREAT GRAINS	NA	NA	NA	NA
brand_feQUAKER 100% NATURAL	0.4745051	0.0297203	15.966	< 2e-16
brand_feQUAKER CAP N CRUNCH CRUNCH BE	NA	NA	NA	NA
brand_feQUAKER LIFE	NA	NA	NA	NA
brand_feGENERAL MILLS BASIC 4	NA	NA	NA	NA
brand_feGENERAL MILLS CINNAMON TST CR	NA	NA	NA	NA
brand_feGENERAL MILLS WHEATIES	NA	NA	NA	NA

(Intercept)	***
p_jmt	***
fiber	***
sugars	***
flavored	***
fortified	***
market_feMassachusetts	***
market_feTexas	***
market_feIllinois	***
market_feIndiana	***
manufacturer_feKELLOGGS	***
manufacturer_fePEPSICO INC	***
manufacturer_feRALCORP HOLDINGS	***
time_fe5	
time_fe6	***
time_fe7	***
time_fe8	*
time_fe9	***

time_fe10	***
time_fe11	***
time_fe12	***
time_fe13	***
time_fe14	***
time_fe15	***
time_fe16	***
time_fe17	***
time_fe18	***
time_fe19	***
brand_feGENERAL MILLS HONEY NUT CHEER	***
brand_feGENERAL MILLS KIX	***
brand_feGENERAL MILLS LUCKY CHARMS	***
brand_feGENERAL MILLS MULTI GRAIN CHE	***
brand_feGENERAL MILLS REESES PUFFS	***
brand_feGENERAL MILLS TOTAL	***
brand_feKELLOGGS APPLE JACKS	***
brand_feKELLOGGS CORN FLAKES	***
brand_feKELLOGGS CORN POPS	***
brand_feKELLOGGS FROOT LOOPS	***
brand_feKELLOGGS FROSTED FLAKES	***
brand_feKELLOGGS FROSTED MINI WHEATS	***
brand_feKELLOGGS RAISIN BRAN	***
brand_feKELLOGGS RICE KRISPIES	***
brand_feKELLOGGS SPECIAL K	***
brand_feKELLOGGS SPECIAL K RED BERRIE	***
brand_fePOST COCOA PEBBLES	***
brand_fePOST FRUITY PEBBLES	***
brand_fePOST GRAPE NUTS	***
brand_fePOST HONEY BUNCHES OF OATS	***
brand_feQUAKER CAP N CRUNCH	***
brand_feQUAKER OATMEAL SQUARES	***
brand_feGENERAL MILLS COCOA PUFFS	***
brand_feGENERAL MILLS GOLDEN GRAHAMS	***
brand_feGENERAL MILLS TOTAL RAISIN BR	***
brand_feGENERAL MILLS TRIX	
brand_feKASHI GO LEAN CRUNCH	***
brand_feKELLOGGS COCOA KRISPIES	***
brand_feKELLOGGS RAISIN BRAN CRUNCH	
brand_fePOST HONEYCOMB	***
brand_fePOST RAISIN BRAN	***
brand_fePOST SELECTS GREAT GRAINS	
brand_feQUAKER 100% NATURAL	***

```

brand_feQUAKER CAP N CRUNCH CRUNCH BE
brand_feQUAKER LIFE
brand_feGENERAL MILLS BASIC 4
brand_feGENERAL MILLS CINNAMON TST CR
brand_feGENERAL MILLS WHEATIES
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Residual standard error: 0.7674 on 63907 degrees of freedom
Multiple R-squared:  0.4594,    Adjusted R-squared:  0.4589
F-statistic: 936.2 on 58 and 63907 DF,  p-value: < 2.2e-16

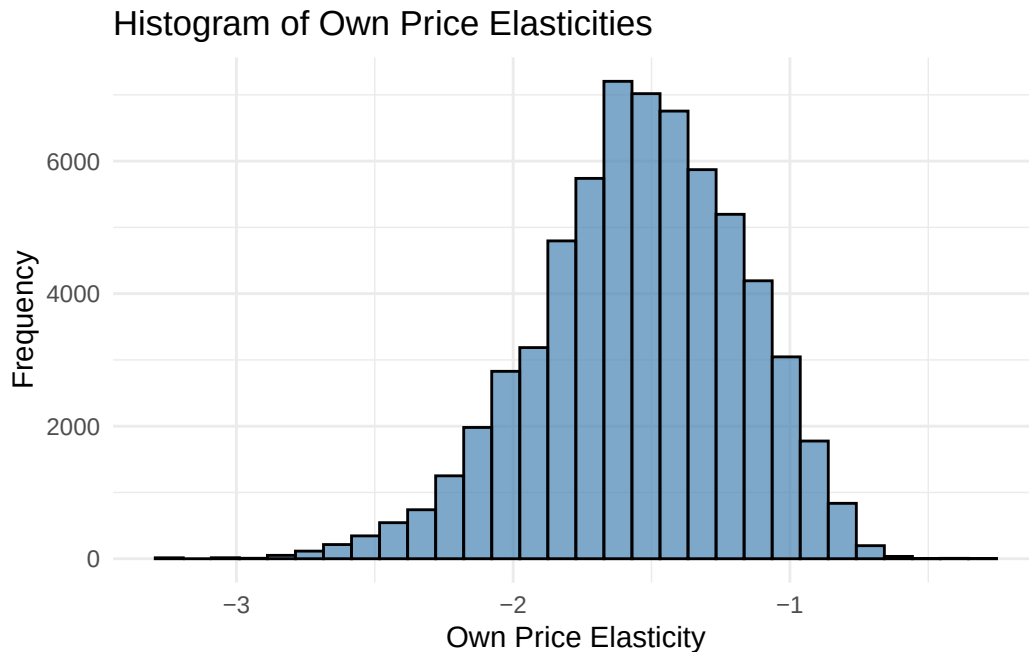
```

```

# Histogram of own price elasticities
reg_data <- reg_data %>%
  mutate(own_price_elasticity = coef(reg_model)["p_jmt"] * p_jmt * (1 - sj))

ggplot(reg_data, aes(x = own_price_elasticity)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "black", alpha = 0.7) +
  labs(title = "Histogram of Own Price Elasticities",
       x = "Own Price Elasticity", y = "Frequency") +
  theme_minimal()

```



2.3

```
# 2SLS with Price Instrument: Price*Quantity of Sugar

reg_data <- reg_data %>%
  mutate(
    z = sugars * sugar_price
  )
stage1 <- lm(p_jmt ~ sugar_price + fiber + sugars + flavored + fortified +
             market_fe + manufacturer_fe + time_fe + brand_fe,
             data = reg_data)
summary(stage1)
```

Call:

```
lm(formula = p_jmt ~ sugar_price + fiber + sugars + flavored +
    fortified + market_fe + manufacturer_fe + time_fe + brand_fe,
    data = reg_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.235706	-0.023291	-0.000373	0.023038	0.244707

Coefficients: (8 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.0644239	0.0214906	2.998	0.002721
sugar_price	0.0007809	0.0001040	7.510	6.00e-14
fiber	0.0225143	0.0008069	27.904	< 2e-16
sugars	0.0052437	0.0002654	19.760	< 2e-16
flavored	-0.0014591	0.0009445	-1.545	0.122394
fortified	-0.0220916	0.0008882	-24.871	< 2e-16
market_feMassachusetts	-0.0163567	0.0004135	-39.555	< 2e-16
market_feTexas	-0.0456712	0.0004755	-96.045	< 2e-16
market_feIllinois	-0.0189938	0.0004882	-38.903	< 2e-16
market_feIndiana	-0.0299661	0.0012992	-23.065	< 2e-16
manufacturer_feKELLOGGS	-0.1352419	0.0029946	-45.162	< 2e-16
manufacturer_fePEPSICO INC	-0.0511758	0.0011190	-45.735	< 2e-16
manufacturer_feRALCORP HOLDINGS	-0.0567750	0.0014453	-39.283	< 2e-16
time_fe5	-0.0084184	0.0011514	-7.311	2.68e-13
time_fe6	-0.0023821	0.0008186	-2.910	0.003617
time_fe7	-0.0087351	0.0011148	-7.835	4.75e-15

time_fe8	-0.0064707	0.0026158	-2.474	0.013376
time_fe9	-0.0114694	0.0022245	-5.156	2.53e-07
time_fe10	-0.0122345	0.0018049	-6.778	1.23e-11
time_fe11	-0.0137985	0.0016547	-8.339	< 2e-16
time_fe12	-0.0053439	0.0007632	-7.002	2.55e-12
time_fe13	0.0016460	0.0016840	0.977	0.328362
time_fe14	0.0009781	0.0016510	0.592	0.553549
time_fe15	0.0002106	0.0019267	0.109	0.912944
time_fe16	-0.0028682	0.0010387	-2.761	0.005758
time_fe17	-0.0089125	0.0008746	-10.191	< 2e-16
time_fe18	-0.0054244	0.0007602	-7.135	9.76e-13
time_fe19	NA	NA	NA	NA
brand_feGENERAL MILLS HONEY NUT CHEER	-0.0479267	0.0016923	-28.320	< 2e-16
brand_feGENERAL MILLS KIX	0.0669586	0.0023612	28.358	< 2e-16
brand_feGENERAL MILLS LUCKY CHARMS	0.0171474	0.0015672	10.942	< 2e-16
brand_feGENERAL MILLS MULTI GRAIN CHE	0.0631274	0.0014372	43.924	< 2e-16
brand_feGENERAL MILLS REESES PUFFS	0.0061599	0.0017462	3.528	0.000419
brand_feGENERAL MILLS TOTAL	0.0183141	0.0025233	7.258	3.97e-13
brand_feKELLOGGS APPLE JACKS	0.0877235	0.0018247	48.075	< 2e-16
brand_feKELLOGGS CORN FLAKES	0.1084513	0.0044924	24.141	< 2e-16
brand_feKELLOGGS CORN POPS	0.0939928	0.0025108	37.436	< 2e-16
brand_feKELLOGGS FROOT LOOPS	0.1206132	0.0019284	62.546	< 2e-16
brand_feKELLOGGS FROSTED FLAKES	0.0675029	0.0025187	26.800	< 2e-16
brand_feKELLOGGS FROSTED MINI WHEATS	0.0167495	0.0017209	9.733	< 2e-16
brand_feKELLOGGS RAISIN BRAN	-0.0366495	0.0016174	-22.659	< 2e-16
brand_feKELLOGGS RICE KRISPIES	0.1978241	0.0045847	43.148	< 2e-16
brand_feKELLOGGS SPECIAL K	0.1459055	0.0039358	37.071	< 2e-16
brand_feKELLOGGS SPECIAL K RED BERRIE	0.1637449	0.0042663	38.381	< 2e-16
brand_fePOST COCOA PEBBLES	0.0640269	0.0019708	32.487	< 2e-16
brand_fePOST FRUITY PEBBLES	0.0590216	0.0019676	29.997	< 2e-16
brand_fePOST GRAPE NUTS	-0.0698779	0.0013403	-52.136	< 2e-16
brand_fePOST HONEY BUNCHES OF OATS	0.0381029	0.0019325	19.717	< 2e-16
brand_feQUAKER CAP N CRUNCH	-0.0140949	0.0014934	-9.438	< 2e-16
brand_feQUAKER OATMEAL SQUARES	-0.0096341	0.0014409	-6.686	2.31e-11
brand_feGENERAL MILLS COCOA PUFFS	0.0037640	0.0017465	2.155	0.031147
brand_feGENERAL MILLS GOLDEN GRAHAMS	0.0277495	0.0013206	21.013	< 2e-16
brand_feGENERAL MILLS TOTAL RAISIN BR	-0.0658601	0.0023868	-27.594	< 2e-16
brand_feGENERAL MILLS TRIX	0.0308858	0.0015860	19.474	< 2e-16
brand_feKASHI GO LEAN CRUNCH	0.1430384	0.0026631	53.711	< 2e-16
brand_feKELLOGGS COCOA KRISPIES	0.1583719	0.0045944	34.471	< 2e-16
brand_feKELLOGGS RAISIN BRAN CRUNCH	NA	NA	NA	NA
brand_fePOST HONEYCOMB	0.0376133	0.0020125	18.690	< 2e-16
brand_fePOST RAISIN BRAN	-0.1396826	0.0034743	-40.204	< 2e-16

brand_fePOST SELECTS GREAT GRAINS	NA	NA	NA	NA
brand_feQUAKER 100% NATURAL	-0.0501302	0.0014514	-34.540	< 2e-16
brand_feQUAKER CAP N CRUNCH CRUNCH BE	NA	NA	NA	NA
brand_feQUAKER LIFE	NA	NA	NA	NA
brand_feGENERAL MILLS BASIC 4	NA	NA	NA	NA
brand_feGENERAL MILLS CINNAMON TST CR	NA	NA	NA	NA
brand_feGENERAL MILLS WHEATIES	NA	NA	NA	NA

(Intercept)	**
sugar_price	***
fiber	***
sugars	***
flavored	
fortified	***
market_feMassachusetts	***
market_feTexas	***
market_feIllinois	***
market_feIndiana	***
manufacturer_feKELLOGGS	***
manufacturer_fePEPSICO INC	***
manufacturer_feRALCORP HOLDINGS	***
time_fe5	***
time_fe6	**
time_fe7	***
time_fe8	*
time_fe9	***
time_fe10	***
time_fe11	***
time_fe12	***
time_fe13	
time_fe14	
time_fe15	
time_fe16	**
time_fe17	***
time_fe18	***
time_fe19	
brand_feGENERAL MILLS HONEY NUT CHEER	***
brand_feGENERAL MILLS KIX	***
brand_feGENERAL MILLS LUCKY CHARMS	***
brand_feGENERAL MILLS MULTI GRAIN CHE	***
brand_feGENERAL MILLS REESES PUFFS	***
brand_feGENERAL MILLS TOTAL	***
brand_feKELLOGGS APPLE JACKS	***

```

brand_feKELLOGGS CORN FLAKES      ***
brand_feKELLOGGS CORN POPS        ***
brand_feKELLOGGS FROOT LOOPS      ***
brand_feKELLOGGS FROSTED FLAKES   ***
brand_feKELLOGGS FROSTED MINI WHEATS ***
brand_feKELLOGGS RAISIN BRAN      ***
brand_feKELLOGGS RICE KRISPIES    ***
brand_feKELLOGGS SPECIAL K        ***
brand_feKELLOGGS SPECIAL K RED BERRIE ***
brand_fePOST COCOA PEBBLES        ***
brand_fePOST FRUITY PEBBLES       ***
brand_fePOST GRAPE NUTS          ***
brand_fePOST HONEY BUNCHES OF OATS ***
brand_feQUAKER CAP N CRUNCH       ***
brand_feQUAKER OATMEAL SQUARES   ***
brand_feGENERAL MILLS COCOA PUFFS  *
brand_feGENERAL MILLS GOLDEN GRAHAMS ***
brand_feGENERAL MILLS TOTAL RAISIN BR ***
brand_feGENERAL MILLS TRIX        ***
brand_feKASHI GO LEAN CRUNCH      ***
brand_feKELLOGGS COCOA KRISPIES   ***
brand_feKELLOGGS RAISIN BRAN CRUNCH
brand_fePOST HONEYCOMB            ***
brand_fePOST RAISIN BRAN          ***
brand_fePOST SELECTS GREAT GRAINS
brand_feQUAKER 100% NATURAL       ***
brand_feQUAKER CAP N CRUNCH CRUNCH BE
brand_feQUAKER LIFE
brand_feGENERAL MILLS BASIC 4
brand_feGENERAL MILLS CINNAMON TST CR
brand_feGENERAL MILLS WHEATIES

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03782 on 63908 degrees of freedom

Multiple R-squared: 0.5348, Adjusted R-squared: 0.5344

F-statistic: 1289 on 57 and 63908 DF, p-value: < 2.2e-16

```

reg_data <- reg_data %>%
  mutate(p_hat = predict(stage1))
stage2 <- lm(u ~ p_hat + fiber + sugars + flavored + fortified +
            market_fe + manufacturer_fe + time_fe + brand_fe,

```

```

      data = reg_data)
summary(stage2)

```

Call:

```

lm(formula = u ~ p_hat + fiber + sugars + flavored + fortified +
    market_fe + manufacturer_fe + time_fe + brand_fe, data = reg_data)

```

Residuals:

Min	1Q	Median	3Q	Max
-4.9432	-0.4490	0.0585	0.5171	3.7097

Coefficients: (8 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.77776	0.17819	9.977	< 2e-16	***
p_hat	-16.10437	0.61805	-26.057	< 2e-16	***
fiber	-0.73116	0.01698	-43.055	< 2e-16	***
sugars	-0.35058	0.00556	-63.057	< 2e-16	***
flavored	0.74597	0.02052	36.350	< 2e-16	***
fortified	0.49411	0.02238	22.076	< 2e-16	***
market_feMassachusetts	-0.26001	0.01338	-19.426	< 2e-16	***
market_feTexas	-0.58427	0.02980	-19.606	< 2e-16	***
market_feIllinois	-0.21735	0.01562	-13.915	< 2e-16	***
market_feIndiana	-0.74870	0.03323	-22.528	< 2e-16	***
manufacturer_feKELLOGGS	2.07206	0.08252	25.109	< 2e-16	***
manufacturer_fePEPSICO INC	0.26683	0.04373	6.102	1.05e-09	***
manufacturer_feRALCORP HOLDINGS	0.92631	0.04419	20.962	< 2e-16	***
time_fe5	-0.08681	0.02666	-3.256	0.001131	**
time_fe6	-0.22904	0.02466	-9.288	< 2e-16	***
time_fe7	-0.19547	0.02829	-6.909	4.93e-12	***
time_fe8	0.10882	0.02779	3.916	9.01e-05	***
time_fe9	-0.10985	0.02518	-4.362	1.29e-05	***
time_fe10	-0.15052	0.02463	-6.111	9.94e-10	***
time_fe11	-0.27179	0.02572	-10.568	< 2e-16	***
time_fe12	-0.33849	0.02574	-13.150	< 2e-16	***
time_fe13	-0.41205	0.02774	-14.851	< 2e-16	***
time_fe14	-0.45882	0.02664	-17.223	< 2e-16	***
time_fe15	-0.43376	0.02779	-15.611	< 2e-16	***
time_fe16	-0.44585	0.02843	-15.680	< 2e-16	***
time_fe17	-0.51518	0.02766	-18.623	< 2e-16	***
time_fe18	-0.58059	0.02607	-22.271	< 2e-16	***
time_fe19	-0.47559	0.02496	-19.054	< 2e-16	***

brand_feGENERAL MILLS HONEY NUT CHEER	2.76326	0.03922	70.451	< 2e-16	***
brand_feGENERAL MILLS KIX	-0.83820	0.05074	-16.521	< 2e-16	***
brand_feGENERAL MILLS LUCKY CHARMS	1.23162	0.03568	34.521	< 2e-16	***
brand_feGENERAL MILLS MULTI GRAIN CHE	1.89829	0.04803	39.526	< 2e-16	***
brand_feGENERAL MILLS REESES PUFFS	0.77843	0.03785	20.567	< 2e-16	***
brand_feGENERAL MILLS TOTAL	1.91560	0.05833	32.838	< 2e-16	***
brand_feKELLOGGS APPLE JACKS	0.20530	0.05709	3.596	0.000323	***
brand_feKELLOGGS CORN FLAKES	-3.16423	0.09101	-34.768	< 2e-16	***
brand_feKELLOGGS CORN POPS	0.12465	0.06046	2.062	0.039252	*
brand_feKELLOGGS FROOT LOOPS	-0.27587	0.07135	-3.867	0.000110	***
brand_feKELLOGGS FROSTED FLAKES	-0.19312	0.04982	-3.876	0.000106	***
brand_feKELLOGGS FROSTED MINI WHEATS	-0.39570	0.03560	-11.116	< 2e-16	***
brand_feKELLOGGS RAISIN BRAN	0.79313	0.04530	17.508	< 2e-16	***
brand_feKELLOGGS RICE KRISPIES	-4.11313	0.12166	-33.809	< 2e-16	***
brand_feKELLOGGS SPECIAL K	-3.00217	0.09809	-30.605	< 2e-16	***
brand_feKELLOGGS SPECIAL K RED BERRIE	-1.93573	0.10362	-18.682	< 2e-16	***
brand_fePOST COCOA PEBBLES	-0.38044	0.05684	-6.693	2.20e-11	***
brand_fePOST FRUITY PEBBLES	0.03393	0.05475	0.620	0.535460	
brand_fePOST GRAPE NUTS	-1.25915	0.05329	-23.630	< 2e-16	***
brand_fePOST HONEY BUNCHES OF OATS	0.55694	0.04592	12.128	< 2e-16	***
brand_feQUAKER CAP N CRUNCH	0.50057	0.03270	15.310	< 2e-16	***
brand_feQUAKER OATMEAL SQUARES	1.28459	0.02913	44.095	< 2e-16	***
brand_feGENERAL MILLS COCOA PUFFS	0.79384	0.03759	21.120	< 2e-16	***
brand_feGENERAL MILLS GOLDEN GRAHAMS	-1.01440	0.03115	-32.565	< 2e-16	***
brand_feGENERAL MILLS TOTAL RAISIN BR	1.00240	0.05944	16.863	< 2e-16	***
brand_feGENERAL MILLS TRIX	0.26047	0.03981	6.542	6.10e-11	***
brand_feKASHI GO LEAN CRUNCH	-1.64681	0.08297	-19.849	< 2e-16	***
brand_feKELLOGGS COCOA KRISPIES	-5.38243	0.10683	-50.384	< 2e-16	***
brand_feKELLOGGS RAISIN BRAN CRUNCH	NA	NA	NA	NA	
brand_fePOST HONEYCOMB	-0.49101	0.05077	-9.672	< 2e-16	***
brand_fePOST RAISIN BRAN	1.11289	0.08779	12.676	< 2e-16	***
brand_fePOST SELECTS GREAT GRAINS	NA	NA	NA	NA	
brand_feQUAKER 100% NATURAL	NA	NA	NA	NA	
brand_feQUAKER CAP N CRUNCH CRUNCH BE	NA	NA	NA	NA	
brand_feQUAKER LIFE	NA	NA	NA	NA	
brand_feGENERAL MILLS BASIC 4	NA	NA	NA	NA	
brand_feGENERAL MILLS CINNAMON TST CR	NA	NA	NA	NA	
brand_feGENERAL MILLS WHEATIES	NA	NA	NA	NA	

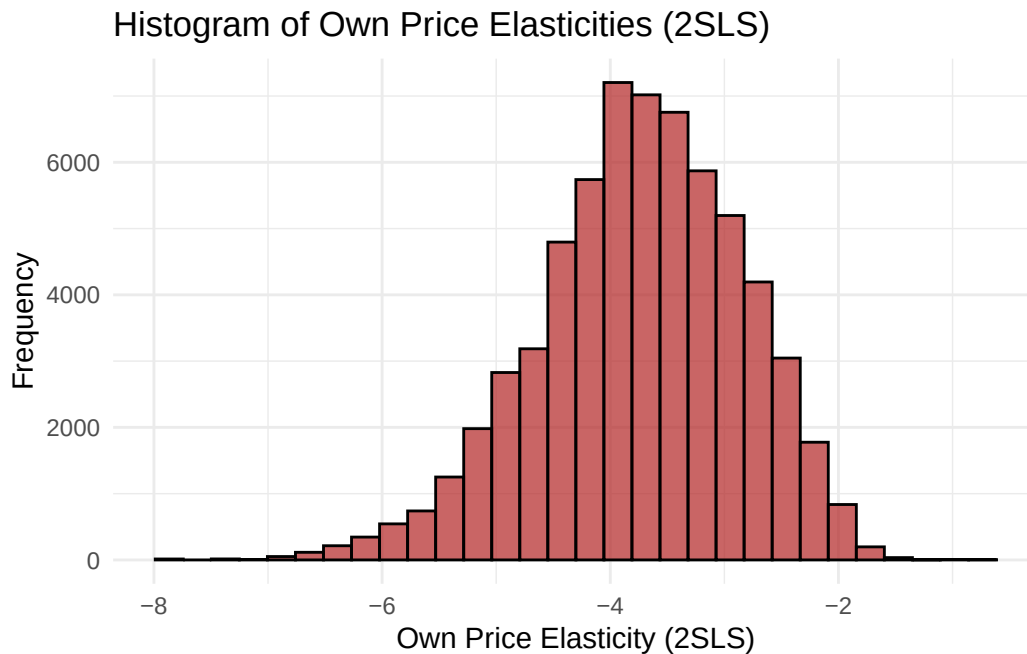
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8074 on 63908 degrees of freedom

Multiple R-squared: 0.4015, Adjusted R-squared: 0.4009

F-statistic: 752.1 on 57 and 63908 DF, p-value: < 2.2e-16

```
# Histogram of own price elasticities (2SLS)
reg_data <- reg_data %>%
  mutate(own_price_elasticity_2sls = coef(stage2)["p_hat"] * p_jmt * (1 - sj))
ggplot(reg_data, aes(x = own_price_elasticity_2sls)) +
  geom_histogram(bins = 30, fill = "firebrick", color = "black", alpha = 0.7) +
  labs(title = "Histogram of Own Price Elasticities (2SLS)",
       x = "Own Price Elasticity (2SLS)", y = "Frequency") +
  theme_minimal()
```



Using 2SLS results in estimates which are more negative than the OLS estimates, indicating that the OLS estimates were biased towards zero.

2.4

```
# Elasticity Matrix for Top 5 Brands by Sales
## Create average matrix by taking average of market level elasticity matrices

top_brands <- reg_data %>%
  group_by(brand) %>%
  summarise(total_sales = sum(quantity, na.rm = TRUE), .groups = "drop") %>%
```

```

  arrange(desc(total_sales)) %>%
  slice_head(n = 5) %>%
  pull(brand)
elasticity_matrices <- list()
for (mkt in unique(reg_data$market_id)) {
  mkt_data <- reg_data %>% filter(market_id == mkt, brand %in% top_brands)
  n <- nrow(mkt_data)
  if (n < 5) next # Skip markets with fewer than 5 top brands
  elas_matrix <- matrix(0, nrow = n, ncol = n)
  rownames(elas_matrix) <- mkt_data$brand
  colnames(elas_matrix) <- mkt_data$brand
  for (i in 1:n) {
    for (j in 1:n) {
      if (i == j) {
        elas_matrix[i, j] <- coef(stage2)["p_hat"] * mkt_data$p_jmt[i] * (1 - mkt_data$sj[i])
      } else {
        elas_matrix[i, j] <- -coef(stage2)["p_hat"] * mkt_data$p_jmt[j] * mkt_data$sj[i]
      }
    }
  }
  elasticity_matrices[[as.character(mkt)]] <- elas_matrix
}
# Average elasticity matrix
avg_elasticity_matrix <- Reduce("+", elasticity_matrices) / length(elasticity_matrices)
avg_elasticity_matrix

```

	KELLOGGS FROSTED MINI WHEATS	
KELLOGGS FROSTED MINI WHEATS		-3.37200991
GENERAL MILLS CHEERIOS		0.06709892
GENERAL MILLS HONEY NUT CHEER		0.06811048
POST HONEY BUNCHES OF OATS		0.06817246
KELLOGGS FROSTED FLAKES		0.06610547
	GENERAL MILLS CHEERIOS	
KELLOGGS FROSTED MINI WHEATS	0.06768931	
GENERAL MILLS CHEERIOS	-3.40399342	
GENERAL MILLS HONEY NUT CHEER	0.06855094	
POST HONEY BUNCHES OF OATS	0.06880160	
KELLOGGS FROSTED FLAKES	0.06682607	
	GENERAL MILLS HONEY NUT CHEER	
KELLOGGS FROSTED MINI WHEATS	0.06801698	
GENERAL MILLS CHEERIOS	0.06772192	
GENERAL MILLS HONEY NUT CHEER	-3.39486250	

POST HONEY BUNCHES OF OATS	0.06905930
KELLOGGS FROSTED FLAKES	0.06609879
POST HONEY BUNCHES OF OATS	
KELLOGGS FROSTED MINI WHEATS	0.06730200
GENERAL MILLS CHEERIOS	0.06764717
GENERAL MILLS HONEY NUT CHEER	0.06835857
POST HONEY BUNCHES OF OATS	-3.39097549
KELLOGGS FROSTED FLAKES	0.06670128
KELLOGGS FROSTED FLAKES	
KELLOGGS FROSTED MINI WHEATS	0.06848859
GENERAL MILLS CHEERIOS	0.06754281
GENERAL MILLS HONEY NUT CHEER	0.06811142
POST HONEY BUNCHES OF OATS	0.06903433
KELLOGGS FROSTED FLAKES	-3.40866766

2.5

```
# Calculaete markups for each manufacturer
reg_data <- reg_data %>%
  mutate(
    own_price_elasticity_2sls = coef(stage2)["p_hat"] * p_jmt * (1 - sj),
    markup = -p_jmt / own_price_elasticity_2sls
  )
markup_summary <- reg_data %>%
  group_by(parent) %>%
  summarise(
    avg_markup = mean(markup, na.rm = TRUE),
    sd_markup = sd(markup, na.rm = TRUE),
    median_markup = median(markup, na.rm = TRUE),
    max_markup = max(markup, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(avg_markup))
markup_summary
```

```
# A tibble: 4 x 5
  parent      avg_markup sd_markup median_markup max_markup
  <chr>      <dbl>      <dbl>      <dbl>      <dbl>
1 RALCORP HOLDINGS 0.0627 0.000891 0.0624 0.0758
2 KELLOGGS        0.0626 0.000520 0.0625 0.0700
```

3 GENERAL MILLS	0.0625	0.000491	0.0623	0.0676
4 PEPSICO INC	0.0624	0.000342	0.0624	0.0698